Decision Making under Uncertainty
Assignment

Fall 2026-2027

Assignment Instructions and Submission Guidelines

We consider a newsvendor problem with unreliable suppliers. For the details on the model, notation, and assumptions, see the article.

Please follow the guidelines below for your assignment submission:

- **File Format:** Submit your report as a PDF file. Include a separate zipped folder containing the source codes used for your analysis and calculations.

- **Deadline:** Ensure that your submission is completed and submitted (via Canvas) by the deadline **(November 30, 2026)**, before midnight.

- **Group Submissions:** The assignment can be worked out by up to **four** students. Only one submission is required from each group.

- **File Name:** Name your report file as "Group # DMUU Assignment.pdf".

- **Peer/Self Evaluation:** Submit the peer/self-evaluation form separately as an individual submission.

If you need further clarification, do not hesitate to reach out to your instructor.

Report Format and quality (2 points)

The assignment will also be graded on the quality of the written report. Marks will reflect clarity of writing, meaningful titles and figures, a well-structured layout.

1. Introduction and Problem Data (1 point)

Present a clear definition of the problem, specifying the parameters and decision variables. Introduce any essential concepts needed to set the stage for your analysis.

2. Solution Algorithm Implementation (2 points)

Present the solution algorithm.

3. Monte Carlo Simulation (8 points)

Introduce the models and context. Present results using tables and figures where appropriate. Provide discussion and interpretation of findings.

4. Stochastic Programming (8 points)

Present your SAA models, algorithms, and results in a clear and structured manner, supported by tables and figures.

5. Risk Averse Models (5 points)

Present the CVaR and VaR models, report numerical results, and discuss managerial insights.

6. Distributionally Robust Optimization (4 points)

Provide your DRO formulation, analysis, and key findings.

7. Conclusion

Summarize the main insights and managerial takeaways from your analysis.

References

List all references and sources cited in your analysis.

Appendices

Include additional material such as large tables, extended numerical results, or supplementary figures.

Weeks	**Suggested Timeline**
Week 1	Form groups
Week 2	Form groups
Week 3	Part 1: Introduction, data generation, heuristic design
Week 4	Part 2: Algorithm implementation and testing
Week 5	Part 3: Monte Carlo Simulation
Week 6	Part 3: Monte Carlo Simulation
Week 7	Part 4: Stochastic Programming
Week 8	Part 4: Stochastic Programming
Week 9	Part 4: Stochastic Programming
Week 10	Part 5: Risk Averse Decision Making
Week 11	Part 6: Distributionally Robust Optimization
Week 12	Part 6: Distributionally Robust Optimization
Week 13	Finalize your results and report
November 30	**Submit the assignment**

1 Introduction and Data (1 point)

In this section, you will introduce the problem setting and construct the data instances used in the rest of the assignment. No optimization results are required in this section. The purpose is to make sure that each group works with a clearly defined benchmark instance and with reproducible randomly generated instances.

Benchmark instance from the article

We use the supplier data in Table 1 as the benchmark instance.

Supplier	p_i	y_i	c_i	c_i^{eff}	CR_i	$\mathbb{E}[R_i]$	$\text{Var}(R_i)$
1	0.05	0.80	40	39.6	0.40	0.99	0.00
2	0.25	0.40	25	21.3	0.70	0.85	0.07
3	0.15	0.50	35	32.4	0.50	0.93	0.03
4	0.10	0.70	30	29.1	0.60	0.97	0.01
5	0.20	0.40	21	18.5	0.78	0.88	0.06
6	0.08	0.60	55	53.2	0.10	0.97	0.01
7	0.12	0.75	45	43.7	0.30	0.97	0.01
8	0.04	0.90	37	36.9	0.46	1.00	0.00
9	0.17	0.65	52	48.9	0.16	0.94	0.02
10	0.22	0.85	42	40.6	0.36	0.97	0.00

Table 1: Supplier data for the benchmark instance. Values are rounded.

The remaining parameters are

$$k = 15, \qquad s = 45, \qquad r = 10.$$

The benchmark demand distribution is

$$D \sim \mathcal{N}(400, 110^2).$$

Random instances

Each group should construct 10 random instances based on the benchmark data. Use the group number as part of the random seed. If g is the group number and $\ell = 1, \ldots, 10$ is the random instance number, use

$$\text{seed}_{g,\ell} = 1000g + \ell.$$

For example, Group 3 should use the seeds $3001, \ldots, 3010$.

For each random instance $\ell = 1, \ldots, 10$, generate new supplier parameters independently as follows:

$$p_{i,\ell} \sim U\left(\max\{0.02, p_i - 0.05\}, \min\{0.35, p_i + 0.05\}\right),$$

$$y_{i,\ell} \sim U\left(\max\{0.30, y_i - 0.15\}, \min\{0.95, y_i + 0.15\}\right),$$

and

$$c_{i,\ell} \sim U\left(\max\{r+1, 0.90c_i\}, \min\{s+k-1, 1.10c_i\}\right).$$

These bounds ensure that

$$r < c_{i,\ell} < s + k,$$

and therefore

$$0 < CR_{i,\ell} < 1.$$

Generate the demand parameters for instance ℓ as follows:

$$\mu_\ell \sim U(350, 450).$$

Then generate the coefficient of variation

$$v_\ell \sim U(0.20, 0.29),$$

and set

$$\sigma_\ell = v_\ell \mu_\ell.$$

Hence,

$$D_\ell \sim \mathcal{N}(\mu_\ell, \sigma_\ell^2),$$

with

$$\frac{\sigma_\ell}{\mu_\ell} = v_\ell < 0.3.$$

For every random instance, recompute the supplier-specific quantities using the generated values:

$$\mathbb{E}[R_{i,\ell}] = 1 - p_{i,\ell}(1 - y_{i,\ell}),$$

$$\mathrm{Var}(R_{i,\ell}) = p_{i,\ell}(1 - p_{i,\ell})(1 - y_{i,\ell})^2,$$

$$c_{i,\ell}^{\text{eff}} = c_{i,\ell}\mathbb{E}[R_{i,\ell}],$$

and

$$CR_{i,\ell} = \frac{s + k - c_{i,\ell}}{s + k - r}.$$

Required output for this section

Each group should recreate Table 1 for the 10 randomly generated instances. For each random instance $\ell = 1, \ldots, 10$, provide a supplier-data table with the generated values

$$p_{i,\ell}, \qquad y_{i,\ell}, \qquad c_{i,\ell},$$

and the recomputed values

$$c_{i,\ell}^{\text{eff}}, \qquad CR_{i,\ell}, \qquad \mathbb{E}[R_{i,\ell}], \qquad \mathrm{Var}(R_{i,\ell}).$$

In addition, for each random instance, report the generated demand parameters

$$\mu_\ell, \qquad \sigma_\ell,$$

and the coefficient of variation

$$v_\ell = \frac{\sigma_\ell}{\mu_\ell}.$$

Use a clear rounding convention, for example three decimal places for probabilities and yield-related quantities, and two decimal places for costs and demand parameters.

Instance selection for the computational sections

Each group should generate all 10 random instances as described above. However, for the computational sections of the assignment, each group should use only two instances:

- the benchmark instance from Table 1;
- one selected random instance, denoted by $\ell^\star$.

The selected random instance $\ell^\star$ should be chosen by the group and used consistently in all later sections. In the report, clearly state which random instance was selected, for example

$$\ell^\star = 4.$$

2 Solution Algorithm Implementation (2 point)

In this part, you will solve the *risk-neutral* version of the problem. The objective is to maximize the firm's expected profit.

You should use the *Preference-Order-Based Supplier Selection Algorithm* provided in Section 5.3 of the article. You do not need to design a new algorithm. Instead, you should explain the optimality properties that lead to the algorithm in your own words and implement the algorithm for the selected data instances.

Use two instances in this section:

- the benchmark instance from Table 1;
- the selected random instance $\ell^\star$.

Tasks.

1. Describe the main idea of the algorithm.

   Briefly explain the optimality properties behind the algorithm. In particular, explain why suppliers are considered in increasing order of procurement cost c_i, how the active, inactive, and remaining supplier sets are updated, and how the order quantities Q_i are determined for the active suppliers.

2. Provide pseudocode.

   You may base your pseudocode on the algorithm in Section 5.3 of the article, but write it clearly using your own notation. The pseudocode should be detailed enough to show the main steps of your implementation.

3. Implement the algorithm for the risk-neutral problem.

   Apply the algorithm to:

   - the benchmark instance;
   - the selected random instance $\ell^\star$.

4. Report the order quantities.

   For each of the two instances, report the order quantity assigned to each supplier:

   $$Q_1, Q_2, \ldots, Q_n.$$

   Suppliers that are not selected should have $Q_i = 0$.

5. Compute the expected profit.

   For each of the two instances, compute the firm's expected profit using the Normal-demand loss function. You may use the standard Normal loss function

   $$L(z) = \phi(z) - z\big(1 - \Phi(z)\big),$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ denote the standard Normal density and distribution functions, respectively.

Clearly report the expected profit for each instance.

6. Compute additional risk measures for the obtained solution.

   For the risk-neutral solution found in each instance, compute the *variance of profit* and the *weighted variance of profit* (see the article for these definitions).

   Report, for each instance:

$$E[G(Q)], \qquad G_V(Q), \qquad G_{WV}(Q).$$

Required output for this section

For this section, each group should submit:

1. A short explanation of the optimality properties behind the Preference-Order-Based Supplier Selection Algorithm.
2. Pseudocode for their implementation of the algorithm.
3. A table reporting, for the benchmark instance and the selected random instance $\ell^\star$:
   - the active suppliers;
   - the order quantities $Q_1, \ldots, Q_n$;
   - the expected profit $E[G(Q)]$;
   - the profit variance $G_V(Q)$;
   - the weighted variance $G_{WV}(Q)$.
4. A short discussion comparing the benchmark instance and the selected random instance. In particular, comment on how supplier costs, disruption probabilities, and yield levels affect supplier selection and order quantities.

3 Monte Carlo Simulation (8 points)

In this section, you will evaluate the solution obtained from the *Preference-Order-Based Supplier Selection Algorithm* using Monte Carlo simulation. The goals are to compare simulated and analytical expected profit, study confidence intervals, apply variance reduction, and test the solution in a volatile out-of-sample environment.

Use only the selected random instance $\ell^\star$. Use the order quantities Q^{alg} obtained in the previous section.

3.1 Raw Simulation and Confidence Intervals (3 points)

Simulation procedure

For each replication $j = 1, \ldots, N$:

1. Generate demand
$$D^{(j)} \sim \mathcal{N}\left(\mu_{\ell^\star}, (\sigma_{\ell^\star})^2\right).$$
Negative demand realizations should be truncated to zero.

2. Generate supplier yields:
$$R_i^{(j)} = \begin{cases} 1, & \text{with probability } 1 - p_{i,\ell^\star}, \\ y_{i,\ell^\star}, & \text{with probability } p_{i,\ell^\star}. \end{cases}$$

3. Compute realized supply:
$$X^{(j)}(Q^{\text{alg}}) = \sum_{i=1}^{n} R_i^{(j)} Q_i^{\text{alg}}.$$

4. Compute the realized profit
$$\Pi^{(j)}(Q^{\text{alg}})$$
using the profit function defined in the article.

Tasks

1. Choose a suitable number of replications N and explain your choice.

2. Report:

   - the simulated mean profit;
   - the standard deviation of simulated profit;
   - a 95% confidence interval for expected profit.
   - a histogram of the simulated profit distribution (and your comments on its shape).

3. Compare your simulated mean profit with the analytical expected profit computed in the previous section.

4. Perform a confidence interval coverage experiment. Repeat the Monte Carlo simulation $M = 100$ times. Each run should use N replications and a different random seed.

   For run $m = 1, \ldots, 100$, construct

$$CI_m = \left[\bar{\Pi}_m - 1.96 \frac{S_m}{\sqrt{N}},\ \bar{\Pi}_m + 1.96 \frac{S_m}{\sqrt{N}} \right],$$

   where $\bar{\Pi}_m$ and S_m are the sample mean and sample standard deviation from run m.

   Using the analytical expected profit as the true mean, report the empirical coverage rate:

$$\text{Coverage rate} = \frac{\#\{m : \text{ analytical expected profit lies in } CI_m\}}{100}.$$

3.2 Variance Reduction for the Algorithmic Solution (2 points)

Apply variance reduction to Q^{alg}. Use the same total number of replications and the same base random seeds as in the plain Monte Carlo simulation.

3.2.1 Control variate (1 point)

Use the realized demand–supply imbalance as the control variate:

$$C^{(j)} = D^{(j)} - X^{(j)}(Q^{\text{alg}}),$$

with

$$E[C] = E[D] - \sum_{i=1}^{n} E[R_i] Q_i^{\text{alg}}.$$

The control-variate estimator is

$$\Pi_{\text{cv}}^{(j)} = \Pi^{(j)}(Q^{\text{alg}}) + \hat{\beta} \left(C^{(j)} - E[C] \right),$$

where

$$\hat{\beta} = -\frac{\widehat{\text{Cov}}(\Pi, C)}{\widehat{\text{Var}}(C)}.$$

Report

$$\widehat{\text{Var}}(\bar{\Pi}) = \frac{S_{\Pi}^2}{N}, \qquad \widehat{\text{Var}}(\bar{\Pi}_{\text{cv}}) = \frac{S_{\text{cv}}^2}{N},$$

and compute

$$\%VR_{\text{cv}} = 100 \frac{\widehat{\text{Var}}(\bar{\Pi}) - \widehat{\text{Var}}(\bar{\Pi}_{\text{cv}})}{\widehat{\text{Var}}(\bar{\Pi})}.$$

Briefly explain why the control variable is related to profit and discuss the variance reduction obtained.

3.2.2 Antithetic demand (1 point)

Apply antithetic variates to demand in the original simulation setting. Do not combine this method with the control variate.

Use the same total number N of model evaluations as in the plain Monte Carlo simulation. Assume that N is even and generate $N/2$ antithetic pairs.

For each pair $j = 1, \ldots, N/2$, draw

$$Z^{(j)} \sim \mathcal{N}(0, 1)$$

and construct

$$D_+^{(j)} = \max\{0, \mu_{\ell^\star} + \sigma_{\ell^\star} Z^{(j)}\},$$

and

$$D_-^{(j)} = \max\{0, \mu_{\ell^\star} - \sigma_{\ell^\star} Z^{(j)}\}.$$

For each of the two demand realizations, independently generate the supplier yields according to the original simulation model.

Using Q^{alg}, compute

$$\Pi_+^{(j)}(Q^{\text{alg}}) \qquad \text{and} \qquad \Pi_-^{(j)}(Q^{\text{alg}}),$$

and define the pair average

$$\widetilde{\Pi}^{(j)} = \frac{1}{2}\left[\Pi_+^{(j)}(Q^{\text{alg}}) + \Pi_-^{(j)}(Q^{\text{alg}})\right].$$

Estimate expected profit by

$$\bar{\Pi}_{\text{anti}} = \frac{1}{N/2} \sum_{j=1}^{N/2} \widetilde{\Pi}^{(j)}.$$

Tasks.

1. Report the estimated expected profit and

$$\widehat{\text{Var}}(\bar{\Pi}_{\text{anti}}) = \frac{S_{\widetilde{\Pi}}^2}{N/2}.$$

2. Compute the percentage variance reduction relative to plain Monte Carlo:

$$\%VR_{\text{anti}} = 100 \frac{\widehat{\text{Var}}(\bar{\Pi}) - \widehat{\text{Var}}(\bar{\Pi}_{\text{anti}})}{\widehat{\text{Var}}(\bar{\Pi})}.$$

3. Discuss whether antithetic sampling reduced the estimator variance and explain your findings in terms of the relationship between demand and realized profit.

3.3 Stress Testing with Correlated Disruptions and a Spot Market (3 points)

In this subsection, evaluate the algorithmic solution Q^{alg} under a more volatile out-of-sample environment. Use the selected random instance $\ell^\star$ and keep Q^{alg} fixed throughout the experiment.

The stress test includes:

- correlated supplier disruptions;
- higher and more volatile demand during disruptions;
- an emergency spot market for covering shortages.

Correlated supplier disruptions

Partition the suppliers into three groups:

$$\mathcal{G}_1 = \{1, 2, 3\}, \qquad \mathcal{G}_2 = \{4, 5, 6\}, \qquad \mathcal{G}_3 = \{7, 8, 9, 10\}.$$

For each simulation replication j, generate

$$E_g^{(j)} \sim \text{Bernoulli}(0.15), \qquad g = 1, 2, 3,$$

independently across groups. Here, $E_g^{(j)} = 1$ means that supplier group g is exposed to a common adverse event that increases the disruption probabilities of the suppliers in that group.

Let

$$H^{(j)} = \sum_{g=1}^{3} E_g^{(j)}$$

denote the number of supplier groups exposed to a common adverse event.

For supplier $i \in \mathcal{G}_g$, define

$$p_i^{(j)} = \begin{cases} p_{i,\ell^\star}, & \text{if } E_g^{(j)} = 0, \\ \min\{0.90, p_{i,\ell^\star} + 0.25\}, & \text{if } E_g^{(j)} = 1. \end{cases}$$

Generate

$$Z_i^{(j)} \sim \text{Bernoulli}\left(p_i^{(j)}\right),$$

independently across suppliers conditional on the group-level events, and define the realized yield factor as

$$R_i^{(j)} = \begin{cases} 1, & \text{if } Z_i^{(j)} = 0, \\ y_{i,\ell^\star}, & \text{if } Z_i^{(j)} = 1. \end{cases}$$

The realized supply is

$$X^{(j)} = \sum_{i=1}^{10} R_i^{(j)} Q_i^{\text{alg}}.$$

Supplier disruptions are positively correlated within each group because the suppliers are affected by the same group-level adverse event. The group-level events are independent across groups.

Demand under stress

Use the demand parameters of the selected random instance:

$$\mu_{\ell^\star}, \qquad \sigma_{\ell^\star} = v_{\ell^\star}\mu_{\ell^\star}.$$

For each replication j, first generate baseline demand:

$$D_0^{(j)} \sim \mathcal{N}\left(\mu_{\ell^\star}, \sigma_{\ell^\star}^2\right).$$

The number of disrupted supplier groups increases demand. Define the stressed demand as

$$D^{(j)} = \max\left\{0,\ \left(1 + 0.10H^{(j)}\right) D_0^{(j)}\right\}.$$

Thus, demand follows approximately the original distribution when no supplier group is disrupted. Each disrupted group increases both the demand mean and standard deviation by 10%. The coefficient of variation remains approximately equal to

$$v_{\ell^\star} = \frac{\sigma_{\ell^\star}}{\mu_{\ell^\star}}.$$

Spot market price

Generate the spot market price conditional on the number of disrupted groups:

$$SP^{(j)} \sim U\left(30 + 15H^{(j)}, 60 + 15H^{(j)}\right).$$

Thus, spot market prices tend to be higher when more supplier groups are disrupted. The shortage before using the spot market is

$$S^{(j)} = \left(D^{(j)} - X^{(j)}\right)^+.$$

The spot market is used only when its unit price is lower than the cost of a lost sale:

$$SP^{(j)} < s + k.$$

Assume that the spot market has sufficient capacity to cover the complete shortage. The spot market purchase quantity is therefore

$$B_{\text{spot}}^{(j)} = S^{(j)}\mathbf{1}\{SP^{(j)} < s + k\}.$$

The remaining lost sales are

$$L^{(j)} = S^{(j)} - B_{\text{spot}}^{(j)}.$$

The realized stress-test profit is $\Pi_{\text{stress}}^{(j)}$.

Tasks

Use at least

$$N_{\text{stress}} = 10{,}000$$

simulation replications and report the random seed.

1. Simulate the stress-testing environment for the fixed solution Q^{alg}.
2. Report the mean profit, profit standard deviation, and a 95% confidence interval for expected profit.
3. Report:

   frequency of spot market use, mean spot market purchase,

   mean spot market cost, mean remaining lost sales.

4. Plot a histogram of the simulated profit distribution.
5. Briefly discuss how correlated disruptions, increased demand, and higher spot market prices affect the profit and risk of Q^{alg}.

4 Stochastic Programming with Spot-Market Recourse (8 points)

In this section, formulate and solve an SAA model for the stress-testing environment using the selected random instance ($\ell^\star$).

The first-stage decisions are the procurement quantities. After demand, supplier yields, and spot prices are realized, the firm may use the spot market to cover shortages when this is cheaper than lost sales.

Generate the SAA scenarios using the same procedure as in the Monte Carlo stress test. Optimize the procurement quantities for this environment and compare the resulting (Q^{SAA})with (Q^{alg})using common random numbers.

4.1 Problem Formulation (1 point)

Formulate the SAA problem as a two-stage expected-profit maximization model using the stress-test scenarios

$$D_\omega, \qquad R_{i\omega}, \qquad SP_\omega, \qquad \omega = 1, \ldots, N.$$

Tasks.

- Define all first-stage and second-stage decision variables.
- State the objective function.
- List and explain the constraints.

4.2 SAA Algorithm (1 point)

Explain how you apply the SAA method to the stress-testing problem for the selected random instance $\ell^\star$.

- Explain your SAA setup.
- Provide pseudo-code for the SAA algorithm, including the steps used to estimate the statistical lower and upper bounds and the 95% confidence bound on the true optimality gap.
- Clearly state the SAA parameters:

$$N, \qquad M, \qquad N',$$

where N is the number of scenarios per SAA batch, M is the number of independent SAA batches, and N' is the number of out-of-sample evaluation scenarios.

- Explain how you select the final SAA solution

$$Q^{\mathrm{SAA}}$$

from the candidate solutions.

Since the objective is profit maximization, be clear about the direction of the bounds you report.

4.3 SAA Analysis (2 points)

Report results for the following SAA settings:

$$(N, M) \in \{(50, 5), (1000, 5), (2000, 5)\}.$$

For each setting, evaluate the candidate SAA solutions using a large independent out-of-sample simulation sample of size

$$N' = 10{,}000.$$

For each setting, report:

- the candidate solution;
- the SAA objective value;
- the out-of-sample estimated expected profit;
- the relevant lower and upper statistical bounds, as described in the slides;
- the estimated 95% confidence bound on the true optimality gap;
- computing time.

Compare the three SAA settings in terms of statistical quality, confidence interval width, solution stability, and runtime.

4.4 SAA Solution and Comparison (2 points)

Using the results from the previous subsection, select the best candidate from

$$Q^{50}, \qquad Q^{1000}, \qquad Q^{2000},$$

and denote it by

$$Q^{\mathrm{SAA}}.$$

Justify your selection based on out-of-sample performance, solution stability, optimality-gap estimates, and runtime.

Compare Q^{SAA} with the earlier algorithmic solution Q^{alg} under the same stress-testing environment. Use common random numbers so that both solutions are evaluated on identical scenarios.

For both solutions, report:

- supplier selection and order quantities;
- mean profit and profit standard deviation;

For each replication j, compute the paired profit difference

$$d_j = \Pi^{(j)}(Q^{\mathrm{SAA}}) - \Pi^{(j)}(Q^{\mathrm{alg}}).$$

Report the mean paired difference and a 95% confidence interval for $E[d_j]$. Conclude whether the SAA solution improves upon the algorithmic solution in the stress-testing environment.

4.5 Value of the Stochastic Solution (1 point)

For the selected random instance, construct the expected-value solution Q^{EV} by replacing demand, supplier yields, and spot price by their expected values.

Evaluate Q^{EV} and the final selected Q^{SAA} on the same large independent out-of-sample sample using common random numbers. Let z_{EEV} and z_{SAA} denote their estimated expected profits.

Compute

$$\mathrm{VSS} = z_{\mathrm{SAA}} - z_{\mathrm{EEV}}.$$

Briefly discuss the value of explicitly modeling uncertainty.

4.6 Value of Perfect Information on Group Disruptions (1 point)

Estimate the value of knowing which supplier groups are exposed to an adverse event before choosing the procurement quantities.

There are $2^3 = 8$ possible group-disruption states

$$e = (E_1, E_2, E_3) \in \{0,1\}^3.$$

For each state e, let

$$H(e) = E_1 + E_2 + E_3.$$

Because the group-level events are independent, the probability of state e is

$$P(e) = 0.15^{H(e)} 0.85^{3-H(e)}.$$

For each state e, assume that the group-level events are known before the procurement quantities are selected. Supplier-level disruptions, demand, and spot-market prices remain uncertain and follow the conditional stress-testing distributions defined earlier.

Solve the conditional stochastic procurement problem for each state e. Evaluate the resulting solution using a large conditional out-of-sample sample, and let z_e^* denote its estimated optimal expected profit. Compute

$$z_{\text{PI-G}} = \sum_{e \in \{0,1\}^3} P(e) z_e^*.$$

The expected value of perfect information on group disruptions is

$$\mathrm{EVPPI}_G = z_{\text{PI-G}} - z_{\text{SAA}},$$

where z_{SAA} is the out-of-sample expected profit of the final SAA solution under the same stress-testing environment.

Briefly discuss the managerial value of receiving advance information about group-level supply disruptions.

5 Risk-Averse Decision Models (5 points)

In this section, extend the stochastic programming model by incorporating risk aversion. Continue working with the selected random instance $\ell^\star$ and the stress-testing environment from the previous sections. Generate one fixed scenario set with $N = 1000$. Use this same scenario set for all models in this section. Let

$$\Pi_\omega(Q)$$

denote the realized profit in scenario ω. Use the same first-stage and second-stage variables and the same recourse constraints as in the SAA model with spot-market recourse.

As a reference solution, first solve the risk-neutral expected-profit model. For the risk neutral and risk averse models report:

- the first-stage order quantities;
- the active suppliers;
- the expected profit;
- the standard deviation of the profit;
- $\text{VaR}_{1-\alpha}$;
- $\text{CVaR}_{1-\alpha}$.

5.1 CVaR Model (2 points)

Formulate and solve a two-stage stochastic program with a CVaR objective. Use a pure CVaR objective, corresponding to $\beta = 1$. Here, α denotes the tail probability. Thus, $\text{CVaR}_{1-\alpha}$ is the average profit in the worst α-fraction of scenarios.

Solve the CVaR model for

$$\alpha \in \{0.01,\ 0.1,\ 0.2,\ 0.4,\ 0.6,\ 0.8,\ 0.9,\ 1\}.$$

5.2 VaR-Constrained Model (2 points)

Formulate and solve a two-stage stochastic program that maximizes expected profit subject to a VaR-type tail constraint. Use an appropriate threshold τ_α.

Solve the VaR-constrained model for

$$\alpha \in \{0.01,\ 0.1,\ 0.2,\ 0.4,\ 0.6,\ 0.8,\ 0.9,\ 1\}.$$

5.3 Comparison (1 point)

Compare the risk-neutral solution with the risk-averse solutions obtained from the CVaR and VaR-constrained models.

Present the results in concise tables. For example, use one table for the CVaR models and one table for the VaR-constrained models:

α	Active suppliers	First-stage orders	Expected profit	Standard dev.	$\text{VaR}_{1-\alpha}$	$\text{CVaR}_{1-\alpha}$
0.01	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
0.1	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
1.0	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

Briefly discuss:

- how the order quantities and active suppliers change as α changes;
- the trade-off between expected profit, profit variance and downside protection;
- the difference between CVaR-based optimization and VaR-constrained optimization;
- feasibility and computational difficulty of the VaR-constrained model;
- which risk-averse policy you would recommend and why.

6 Distributionally Robust Optimization (4 points)

In this section, we continue with the original unreliable-supplier newsvendor setting introduced at the beginning of the assignment and in the reference paper. There is no spot market and no correlated stress-test setting in this section.

Supplier uncertainty is modeled explicitly by enumerating all supplier-yield scenarios. Since each supplier has two possible yield states, there are 2^n supplier-yield scenarios. In this assignment, $n = 10$, so enumeration gives $2^{10} = 1024$ *supplier-yield scenarios*, which is computationally tractable. Therefore, supplier uncertainty is fully represented by a finite scenario set. The distributionally robust part of this section concerns only the demand distribution.

Supplier i has random yield factor

$$R_i = \begin{cases} 1, & \text{with probability } 1 - p_i, \\ y_i, & \text{with probability } p_i. \end{cases}$$

For each *supplier-yield scenario* u, let R_{iu} denote the realized yield factor of supplier i, and let p_u denote the probability of scenario u. The realized total supply in scenario u is

$$X_u(Q) = \sum_{i=1}^{n} R_{iu} Q_i.$$

For a realization of demand D and supplier-yield scenario u, profit is

$$\Pi_u(Q; D) = s \min\{D, X_u(Q)\} + r(X_u(Q) - D)^+ - k(D - X_u(Q))^+ - \sum_{i=1}^{n} c_i R_{iu} Q_i.$$

The objective in the stochastic newsvendor model is

$$\max_{Q_i \geq 0} \mathbb{E}_{D,u} \left[\Pi_u(Q; D) \right].$$

In the earlier sections, the demand distribution was specified. In this section, the demand distribution is only partially specified through moment information. This motivates a distributionally robust optimization (DRO) approach.

Use the selected random instance $\ell^\star$ throughout this section. For the demand moments, use

$$\mu^X = \mu_{\ell^\star}, \qquad \sigma^X = \sigma_{\ell^\star}.$$

6.1 Deterministic Demand (SLP) (0.5 points)

First assume that demand is deterministic:

$$D = \mu^X = \mu_{\ell^\star}.$$

Using the supplier-yield scenarios defined above, formulate the resulting model as a two-stage stochastic linear program (SLP). The first-stage decisions are the order quantities Q_i. The second-stage quantities should represent realized sales, leftover inventory,

and lost sales in each supplier-yield scenario. Solve the linear model and report the optimal order quantities

$$Q_1, \ldots, Q_n.$$

6.2 Mean–MAD Ambiguity (DRO-LP) (1.5 points)

Now assume that demand is uncertain, but its distribution is not fully known. Consider the ambiguity set

$$\mathcal{P}(\mu^X, d^X) = \left\{ \mathbb{P} : \mathrm{supp}(D) \subseteq [0, b^X], \quad \mathbb{E}[D] = \mu^X, \quad \mathbb{E}[|D - \mu^X|] = d^X \right\}.$$

Use

$$\mu^X = \mu_{\ell^\star}, \qquad d^X = \sqrt{\frac{2}{\pi}}\sigma_{\ell^\star}, \qquad b^X = 2\mu_{\ell^\star}.$$

Before applying the mean–MAD extremal distribution result, verify that for fixed order quantities Q and a fixed supplier-yield scenario u, the function

$$-\Pi_u(Q; D)$$

is convex in demand D.

Then use the mean–MAD result for convex functions to construct the worst-case demand distribution. In particular, the worst-case distribution is supported on

$$\{0, \mu^X, b^X\}.$$

The probabilities should be chosen to satisfy

$$\mathbb{E}[D] = \mu^X, \qquad \mathbb{E}[|D - \mu^X|] = d^X.$$

Formulate the corresponding DRO model:

$$\max_{Q_i \geq 0} \inf_{\mathbb{P} \in \mathcal{P}(\mu^X, d^X)} \mathbb{E}_{\mathbb{P},u} \left[\Pi_u(Q; D) \right].$$

Combine the three demand support points with the $2^{10} = 1024$ supplier-yield scenarios and formulate the resulting DRO-LP.

Solve the DRO-LP and report:

- a short explanation showing why $-\Pi_u(Q; D)$ is convex in D, for fixed Q and supplier-yield scenario u;
- the worst-case demand distribution;
- the optimal order quantities $Q_1, \ldots, Q_n$;
- the worst-case expected profit;
- the solver runtime.

6.3 Mean–Variance Ambiguity: Scenario-Wise Scarf Approximation (1.5 points)

Assume that the demand distribution is unknown, but

$$\mathbb{E}[D] = \mu^X, \qquad \mathrm{Var}(D) = (\sigma^X)^2,$$

where

$$\mu^X = \mu_{\ell^\star}, \qquad \sigma^X = \sigma_{\ell^\star}.$$

Supplier uncertainty is represented by the $2^{10} = 1024$ supplier-yield scenarios. For scenario u, define the realized supply as

$$X_u(Q) = \sum_{i=1}^{n} R_{iu} Q_i$$

and the shortage as

$$L_u(Q, D) = (D - X_u(Q))^+ .$$

For a fixed supply level $X_u(Q)$, Scarf's worst-case expected shortage is

$$\psi(X_u(Q)) = \frac{1}{2}\left(\mu^X - X_u(Q) + \sqrt{(X_u(Q) - \mu^X)^2 + (\sigma^X)^2}\right).$$

Apply this formula separately to each supplier-yield scenario and use

$$\sum_u p_u \psi(X_u(Q))$$

as a conservative approximation of the joint worst-case expected shortage. This approximation may be conservative because it may correspond to a different worst-case demand distribution for each supplier-yield scenario.

Formulate and solve the resulting second-order cone program (SOCP). Report:

- the SOCP formulation;
- the optimal order quantities $Q_1, \ldots, Q_n$;
- the conservative worst-case profit bound;
- the solver runtime.

6.4 Comparison (0.5 points)

Compare the deterministic, mean–MAD DRO, and mean–variance DRO solutions.

Present a concise table:

Model	Order quantities	Profit measure	Runtime
Deterministic SLP	$\cdots$	Expected profit	$\cdots$
Mean–MAD DRO-LP	$\cdots$	Worst-case profit	$\cdots$
Mean–variance Scarf SOCP	$\cdots$	Conservative worst-case profit bound	$\cdots$

Briefly discuss:

- how the order quantities change across the three models;
- how the ambiguity set affects the procurement policy;
- whether the DRO solutions are more conservative than the deterministic solution;
- the computational trade-off between the LP and SOCP formulations.

Computational note. For both DRO models, full enumeration of the $2^{10} = 1024$ supplier-yield scenarios is expected to be computationally manageable. If full enumeration is not possible, use a sampled subset of supplier-yield scenarios and clearly report the sample size and sampling procedure.